

Aptimizer Calculation Reference — Corrections

Four corrections, found by checking the document against the code it transcribes — the check page 1 of the reference asks a reader to make.

None of them is a wrong number. 82 constants across pages 3 to 15, and every formula spot-checked, match the source exactly: Rankine, $P_{\text{service}} = P_u / 1.5$, $B = \sqrt{A_{\text{req}}} + 0.05$, the IS 10262 coarse aggregate adjustment, the three Sa/g branches, the cost adder sequence, the S-curve and the IRR bounds. Two things that looked like defects are not: `math.sin` is fed `math.radians(soil[phi])`, and the `min(..., 2.5)` cap on the design spectrum is defensive and can never bind, because the hyperbolic branch begins exactly at 2.5.

1. Page 4, Module 2 — the fundamental period is understated

The document gives the two expressions separately. The code returns the **shorter of the two** for infilled and wall systems, which is the path that runs by default: `frame_type` defaults to `brick_infill`. A reader hand-checking base shear against the document would use the wrong period. Replace both rows with three.

Quantity	Formula as implemented	Note
Fundamental period, bare frame	$T_a = 0.075 H^{0.75}$	IS 1893 Cl. 7.6.2(a). Used as given only when <code>frame_type</code> is <code>bare_frame</code> .
Fundamental period, infilled frame	$T_a = \min(0.075 H^{0.75}, 0.09 H / \sqrt{d})$	Cl. 7.6.2(b); d = base dimension, taken as <code>sqrt(footprint)</code> . The shorter period governs, because a shorter period gives the larger design base shear. This is the default path.
Fundamental period, wall system	$T_a = \min(0.075 H^{0.75}, 0.09 H / \sqrt{d})$	Cl. 7.6.2(d) is not implemented - see Known Simplifications. The output labels this result "wall system approximation".

Source: `iscodes.seismic_period`.

2. Page 16 — a simplification is missing

The code declares this one in its own docstring and labels it in the returned result, but page 16 does not list it. It belongs there, because it is precisely the class of thing that page exists to declare before an examiner finds it. Add:

Shear-wall fundamental period. IS 1893 Cl. 7.6.2(d) gives the period of a shear wall system in terms of the wall area A_w . The platform does not model shear wall geometry, so it substitutes the infilled-frame expression and reports the result as a wall system approximation. The substitution is not guaranteed to be conservative for a wall-dominant building; check the period before relying on the base shear it produces.

3. Page 5, Module 3 — q is undefined

The Rankine row uses q without saying what it is. It is not the SBC, and the definition is not something a reader would reconstruct. Replace the Note column of "Minimum depth (Rankine)" with:

Minimum 0.5 m. $q = \min(\text{SBC}, 4 \times \text{column load} / (\text{bay}_x \times \text{bay}_y))$ - the lesser of the soil's safe bearing capacity and the pressure the column grid imposes. ϕ is converted to radians before use.

Source: `engineering.m3_foundation`.

4. Pages 7 and 10 — the fire figures are now version-keyed

Since the code-version work the three NBC fire constants are no longer unconditional. They are keyed on the project's `code_version`, which defaults to `nbc2016`. The document's figures remain correct **for that default** - but it no longer says so, and that qualification is the entire point of the change.

Page 7, "Constants used"

Constant	Value	Source
High-rise trigger	15 m (NBC 2016)	NBC Part 4. Held in <code>FIRE_WATER_BY_VERSION</code> , not as a literal - this is the value SP 7:2026 is most reported to move.

Fire reserve in sump	25 000 L above the trigger (NBC 2016)	NBC Part 4. Under sp7_2026 this is UNREAD and no reserve is added to the sump.
Static fire storage	50 000 L above the trigger (NBC 2016)	NBC Part 4 Table 7. Same version key.

Page 7, "Sump volume" row — Note becomes

$V_{\text{sump}} = Q + \text{fire reserve}$. The reserve is 25 000 L above the high-rise trigger under NBC 2016. When the active code version has not been read, nothing is added and the output says so, rather than reporting a sump as though the reserve were zero.

Page 10, "Static water" row — Note becomes

NBC Table 7, under NBC 2016. Under SP 7:2026 the module refuses to check rather than reporting a figure.

Page 10, "Constants used" — add a row

Constant	Value	Source
code_version	nbc2016 (default)	FIRE_BY_VERSION. Under sp7_2026 all ten fire thresholds are UNREAD, the module returns unchecked with no score, and it names the values it could not read.

One thing to recount

Page 1 states "14 modules, 106 formulas and 72 constants". The constant count has moved with the version tables. Recount it from the source rather than adjusting it by hand - that number is the sort of thing this document exists to keep honest.